

# Getting the Distribution Is Not Getting the Correspondence

Notation, bisimulation, and the bootstrap problem for grounded symbol systems

6 August 2026

## Abstract

A language model trained to predict the next token inherits a correspondence between symbols and world that it did not construct and cannot inspect. That correspondence was bootstrapped by the human agents who use the language, and it lives in them, not in the text. Undeciphered scripts make the point sharply: we possess the distribution and lack the correspondence, and no amount of the former yields the latter. The current work on sperm whale codas supplies the complementary control: no cognates, no anchor language, and decoding is nonetheless thought feasible—because the population is alive and can be watched using its symbols while it acts. Those same two species are the proof that the problem is soluble: it has been solved twice, independently, and the two solutions share three features—a population rather than an individual, an interface that co-evolved with the symbols rather than being chosen, and a criterion external to the symbol system—which constrain the shape of any third. This note lays out the resulting bootstrap problem, provisionally adopts Nelson Goodman’s theory of notationality as an answer to the prior question of what a symbol system *is*, records where that theory agrees with a bisimulation-based account of symbol identity and where it demands strictly more, and examines the flexibility of the reflective higher-order calculus to admit builtin processes drawn from a host world. Quotation and dereference in the pure calculus turn out to be recursive score construction rather than a score / performance relation; adjoining builtins is precisely what re-types them into one. That extension is developed as far as a calculus-making functor  $\text{RHO}[\mathcal{B}]$  and then deliberately halted: it is offered as a suggestive analogy for what a grounded symbol system would have to look like, not as a solution to the problem of acquiring one. The note closes with an inventory of open questions, several of which appear tractable.

## 1 Introduction

This note is a framing document. It does not report a result; it tries to state a problem precisely enough that results become possible, and to say honestly which of the apparatus already in hand bears on it and which does not.

The occasion is a distinction that has become easy to lose. Transformer networks are applied to two quite different tasks under one description. In the first, the network is a *transducer*: it receives sensor data and must produce symbols. In the second, it is a *next-token predictor*: it receives symbols and must produce more of them. The second task looks like the first performed at scale, and much of the current discourse treats it that way. It is not. In the second case the transduction has already happened, performed over millennia by a population of human agents, and the network is working downstream of a correspondence it neither built nor has access to.

The argument runs in four movements.

- (1) **The problem** (§2). A living language exists inside a correspondence between language and environment, and that correspondence is a fact about the agents, not about the language. Linear A and Rongorongo are the controlled experiment: the distribution survives, the correspondence does not, and the distribution does not recover it.
- (2) **The proposal** (§4). We provisionally adopt Goodman’s notationality as an account of what a symbol system is. It does not solve the bootstrap problem. It says what would count as having solved it, which is a different and prerequisite service.
- (3) **The fit** (§5). Goodman’s conditions align closely with a bisimulation-based treatment of symbol identity, and we say exactly where. Two of his five conditions are quotient conditions that bisimulation supplies. Two are separation conditions that it does not. The gap is the interesting part.
- (4) **The analogy** (§6). In the reflective higher-order calculus, quotation and dereference are mutually inverse but purely score-internal: they witness a faithful recursion inside the notation, with no performance in the picture. Adjoining builtin processes from a host world re-types those same operators into a score / performance correspondence. Read as a calculus-making functor, the extension exhibits several of Goodman’s conditions as structural properties. We develop the reading and then argue explicitly why it is an analogy and not a solution.

Open questions are collected in §9, and §10 separates what is claimed here from what is not.

**Relation to the other notes.** This note is a sibling of *The Mortal Scientist* and *Learning to Play*, and depends on *Composing Learners* for the argument of §7. Its relation to those notes is adversarial in a productive way: it identifies a problem the mortal-scientist framework does not face, explains why it does not face it, and asks what would have to change for it to.

## 2 The problem

### 2.1 A language presupposes a correspondence it does not contain

Consider what a competent speaker has that a corpus does not. The speaker can be handed an object and asked whether it is a coconut. The corpus cannot. The speaker’s competence includes a set of dispositions connecting expressions to features of an environment accessible to the speaker; the corpus records only the traces those dispositions leave in speech. Harnad’s diagnosis remains the clearest: symbolic representations must be grounded in nonsymbolic ones—iconic representations that are analogues of sensory projections, and categorical representations that pick out invariants—and elementary symbols are the names of the categories so picked out [14]. The names are recoverable from text. The categories are not.

Bender and Koller’s octopus makes the same point as a thought experiment, and Bisk et al. generalise it into a hierarchy of “world scopes” along which a text-only system occupies the narrowest rung [4, 7]. Both are usually read as arguments about *understanding*, which invites the reply that understanding is underdefined. We take the sharper reading, which is not about understanding at all: the correspondence is a physical relation between agent and environment, it is a fact about the agent, and text is downstream of it. Whether or not anything deserves to be called understanding, the correspondence is either instantiated somewhere or it is not.

**Observation 2.1** (The locus of correspondence). *For a living language  $L$  used by a population  $A$  in an environment  $E$ , the correspondence  $c : L \rightarrow E$  is realised in  $A$ . The corpus is the image of language use under  $c$ ’s having already been applied; it does not contain  $c$ .*

The observation is trivial when stated. Its force is that the loss of  $A$  destroys  $c$  while leaving the corpus intact, and that this is not a hypothetical.

## 2.2 The evidence: undeciphered scripts

The undeciphered scripts are the controlled experiment for Observation 2.1. Rongorongo and Linear A are corpora without populations. We have the marks. We can compute over them. We do not have the correspondence, and centuries of effort have not manufactured one from the marks alone.

The machine-learning record is unusually informative here, because it isolates the variable. Luo, Cao and Barzilay’s neural decipherment recovers a substantial fraction of Linear B cognates and improves the state of the art on Ugaritic [15]. The method works because Linear B was already known to encode an early form of Greek and Ugaritic an early form of Hebrew. The paradigm is explicitly *cognate-based*: it presupposes an anchor language whose correspondence to the world is intact, and transports the unknown script into that anchor. Follow-up work states the assumption directly and observes that it fails for many undeciphered scripts, the first casualty being knowledge of the language family [16]. Iberian is the standing counterexample; Linear A and Rongorongo are the standing failures.

What the anchor supplies is not vocabulary. It is access to a correspondence that is instantiated somewhere: in the speakers of the anchor language, or in the reconstructed dispositions of their ancestors. The unknown script is transported into a system where  $c$  of Observation 2.1 still exists. That is the only route by which the marks have ever been made to mean anything.

Or rather, it is one of two routes.

## 2.3 The other route: living populations and whale song

The current work on cetacean communication is the cleanest available control on the argument, because it removes the anchor and keeps the population.

Project CETI has no cognates. Sperm whale codas have no known relative in any grounded symbol system, and there is no Greek to transport them into. What the project does have is a living population that can be observed while it vocalises. Analysis of roughly nine thousand codas from the Eastern Caribbean clan yielded a combinatorial coding system with context-sensitive structure—rhythm, tempo, and what the authors term rubato and ornamentation—and a proposed inventory of coda types [22]. Subsequent phonological analysis finds that coda qualities pattern like human vowels along several linguistic dimensions, not merely resemble them acoustically [5].

The important point is what all this structure did *not* deliver, and what the researchers said about it at the time. Sharma, reporting the phonetic-alphabet result, observed that they do not yet know what the whales are saying, and that the next step is to study the calls in their behavioural contexts in order to find out. That sentence is the thesis of this note stated by a practitioner in the field. Increasingly sophisticated distributional analysis has kept yielding more structure and no meaning; the meaning is expected to come from watching the animals use the symbols while doing things.

This is what makes the case decisive rather than merely suggestive. Whale song is exactly Rongorongo minus the extinction. Same absence of cognates, same absence of a bilingual, same abundance of signal. The difference is that the agents are still alive and still exercising the correspondence in observable behaviour, and that difference is the entire reason decoding is thought to be possible at all. The whales are demonstrating  $c$  for the researchers.

**Observation 2.2** (Two routes, neither distributional). *Correspondence has been recovered for an unknown symbol system by exactly two routes: transport into a system whose correspondence is already instantiated in a living or reconstructible population, and direct observation of the using population exercising the correspondence in context. Both routes acquire  $c$  from agents who have it. No correspondence has been constructed from distributional structure alone.*

Rongorongo and Linear A have neither route available. Linear B had the first. The sperm whales offer the second, and the field’s expectations track that availability precisely: nobody expects to decode Linear A by collecting more tablets, and everybody expects to decode codas by collecting more behaviour.

Observation 2.2 is an empirical generalisation over a small sample, and one should not over-read it. But it is a sample with a clear structural explanation, and it is the correct evidential base for a claim otherwise argued purely from intuition pumps.

## 2.4 What next-token prediction does recover

It would be a mistake to conclude that next-token prediction recovers nothing structural. It recovers a great deal, and being precise about *what* sharpens the problem rather than blunting it.

Computational mechanics gives the right vocabulary. For a stochastic process, the *causal states* are the equivalence classes of pasts under identity of conditional future; the resulting  $\epsilon$ -machine is the minimal representation consistent with optimal prediction [21]. The *mixed-state presentation* describes how an optimal observer’s belief over the generator’s hidden states evolves as tokens arrive. Shai et al. show that transformers trained on next-token prediction linearly represent this belief-state geometry in their residual stream, including cases where the geometry is fractal, and that the represented belief states carry information about the entire future rather than only the next token [20].

This is a strong result and it says exactly the wrong thing for the optimist. The structure recovered is the structure of *the process that generated the tokens*. For a corpus of Rongorongo, that process is the scribal community: its conventions, its genres, its habits of repetition. A transformer trained on a sufficiently large Rongorongo corpus would represent the belief-state geometry of Rongorongo scribes. It would not represent the island. The generator is the population, and the population’s correspondence to its environment is not in the token stream.

**Observation 2.3** (The generator is the wrong object). *Next-token prediction recovers the mixed-state presentation of the token-generating process. Where that process is a linguistic community, the recovered structure is the structure of the community’s usage, not of the environment the community’s language corresponds to. The two coincide only to the extent that usage is a faithful transcript of environmental structure, which is precisely what is at issue.*

## 2.5 The bootstrap problem

We can now state the problem this note exists to frame.

**The bootstrap problem.** Given an agent with an interface to an environment, and no antecedently grounded symbol system to transport into, construct a symbol system whose correspondence to that environment is instantiated in the agent.

Three features of the statement matter. First, “instantiated in the agent” rather than “correct”: we are asking for the correspondence to exist and be exercisable, not for it to be

true in some external sense. Second, “no antecedently grounded system” rules out both routes of Observation 2.2: there is no anchor to transport into, and no population already exercising the correspondence to observe. These are the only strategies known to work and both are unavailable by hypothesis. Third, the problem is stated for a single agent, and §7 argues that this is a defect of the statement rather than of the solution.

## 2.6 The problem is soluble

It is worth saying immediately, because it is easy to lose under the weight of the negative results above: the bootstrap problem is not open in the sense that its solubility is in doubt. It has been solved.

Humans solved it. There was no anchor language to transport into and no prior population exercising a correspondence to observe, because ours was the first. Whatever else is unclear, the correspondence  $c$  of Observation 2.1 exists, and it was constructed by agents starting without one.

Sperm whales solved it too, and this is the more informative of the two proofs, because it is independent. The cetacean lineage is dramatically divergent from ours, and the coda system is not a variant of anything human; the Project CETI work is remarkable precisely because it finds combinatorial and contextual organisation of a kind previously thought proprietary to human language in a system that could not have inherited it [22, 5]. Two independent solutions are much stronger evidence than one that the problem has structure making it soluble, rather than that one lineage was lucky.

**Observation 2.4** (Existence). *The bootstrap problem has at least two independent solutions in nature. Any argument that it is insoluble is an argument against an observed fact and is therefore unsound; the question is not whether but how.*

The existence proofs do more than license optimism, because they are not silent about method. Three features are common to both solutions and none of them is incidental.

- (i) **The solver was a population.** Neither humans nor whales bootstrapped as isolated individuals. In both cases the correspondence is carried by a group and transmitted within it, and in both cases the symbol system is a shared object no member constructed alone. This is the same conclusion §7 reaches by an entirely different route, from relabelling-invariance. Two independent arguments converging on the same structural requirement is the strongest signal in this note.
- (ii) **The interface was not chosen.** The set of distinctions available to be symbolised—what §6 calls  $\mathcal{B}$ —was inherited, and was shaped by the same selective process that shaped the use of the symbols. Sensorium and symbol system co-evolved. The extensibility question that we are forced to ask as a design problem was answered in the biological cases by selection on the interface itself.
- (iii) **There was an external criterion.** Both solutions were produced under pressure from something outside the symbol system—survival, reproduction, coordination under threat—which supplied the fitness signal by which one correspondence beat another. Nothing internal to a notational system selects among the relabellings of it, and in the natural cases nothing internal had to.

What is therefore established is that the problem is soluble *by populations, under selection, over long timescales, with co-evolving interfaces*. What is not established is that it is soluble by construction, on demand, for an agent whose interface is fixed by design and whose criterion is supplied by us. The relation is that of flight to birds: the existence proof was decisive and

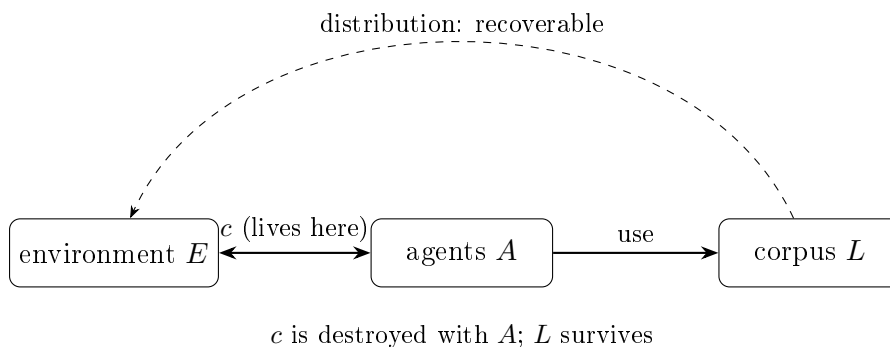


Figure 1: The correspondence  $c$  is a relation between environment and agents. The corpus is downstream. Recovering the distribution over  $L$  does not recover  $c$ , and the dashed arrow is not a reconstruction of  $c$  but a statistic of its traces.

highly informative about constraints, it ruled out a large class of impossibility arguments, and the eventual artefact was not a bird. This note takes the existence proofs in that spirit—as evidence about the shape of the answer, and as a standing rebuke to any framework that makes the answer impossible.

### 3 Why the mortal scientist does not face this problem

The mortal-scientist framework models a learner as a cost-accounted computation in the reflective higher-order calculus, with hypotheses drawn from a namespace logic whose formulae denote bisimulation equivalence classes.<sup>1</sup> It is worth being clear about why that framework does not encounter the bootstrap problem, because the reason is instructive and is not that the problem was solved.

Computations are ontologically isolated, so the environment of a computation is other computations. Both sides of the cut are therefore objects of the same calculus. The hypothesis language is a logic for that calculus, and its formulae denote equivalence classes of exactly the things on the far side of the cut. The correspondence between symbol and referent is consequently *analytic*: there is no gap to bridge because the ontology and the hypothesis language are two presentations of one object. The scientist’s grounding is perfect in the limit for the same reason that a map of a map is accurate—the territory was chosen to be of the same kind as the map.

**Observation 3.1** (The sidestep). *The mortal scientist’s grounding is guaranteed by the coincidence of the model of computation with the model of the environment. Every result of that framework which relies on both sides of the cut being rho-processes requires re-derivation when the far side is not.*

This is not a criticism of the framework, which was built for a different purpose and does that job. It is a statement of scope. The moment we ask for a mind that acts in the world humans act in, the far side of the cut stops being a rho-process and Observation 3.1 takes effect. Everything in §§4–6 is an attempt to say what would have to be added.

<sup>1</sup>The calculus is written rho, never  $\rho$ . The name is an acronym for *reflective higher order*; that it reads as the Greek letter following  $\pi$  is a pun on the rho calculus following the  $\pi$ -calculus, not an assertion that the letter is the name.

## 4 Goodman’s framework, provisionally adopted

### 4.1 Why this framework

The bootstrap problem asks us to construct a symbol system. It is therefore prior to ask what one is. The literature reviewed above is remarkably thin on this: symbol grounding work generally treats “symbol” as understood and asks how to attach it to sensors; representation learning treats discreteness as an architectural choice; emergent communication treats a symbol as whatever the channel carries. None of these gives a criterion by which a proposed system could be judged inadequate.

Goodman’s *Languages of Art* does [13]. It offers a set of conditions distinguishing notational systems from other symbol systems, developed for the score / performance relation in music and dance and generalised from there. We adopt it provisionally—provisionally because it has known failure modes, discussed in §4.4, and because adopting it is a bet that its conditions are the right ones rather than merely the first available ones.

The one prior attempt to put Goodman’s theory to computational work is Goel’s, which derives from notationality a class of *physical notational systems* and argues that these are exactly the dynamical systems capable of realising computational information processing [12]. That work predates the current setting and has not been revisited for learned representations. It is, so far as we can determine, the whole of the prior art.

### 4.2 The conditions

Goodman’s apparatus distinguishes *marks* (physical inscriptions or utterances), *characters* (classes of marks that count as the same), and *compliance* (the things a character applies to, its *compliance class*). A score is a character; a performance is a compliant.

A *notational scheme* satisfies two syntactic conditions:

**Condition 4.1** (Syntactic disjointness). No mark belongs to more than one character.

**Condition 4.2** (Syntactic finite differentiation). For any two characters  $K \neq K'$  and any mark  $m$  not belonging to both, it is theoretically possible to determine that  $m$  does not belong to  $K$ , or that it does not belong to  $K'$ .

A *notational system* is a scheme satisfying three further semantic conditions:

**Condition 4.3** (Unambiguity). Each character has a single compliance class: not different compliants at different times or in different contexts.

**Condition 4.4** (Semantic disjointness). No two characters share a compliant.

**Condition 4.5** (Semantic finite differentiation). The analogue of Condition 4.2 on the compliance side.

The two finite-differentiation conditions are what exclude *dense* systems. A system whose characters are ordered so that between any two there is a third fails Condition 4.2, and this is Goodman’s account of the difference between a score and a sketch, or between digital and analogue.

### 4.3 The roundtrip is derived, not primitive

It is natural—and was our own first reading—to take the criterion of adequacy to be losslessness of the roundtrip: from score to performance and back by transcription, and from performance to score and back, without loss of identity. That is indeed the property Goodman is after, and it is the property that matters for our purposes. But it is a consequence

of Conditions 4.1–4.5, not an additional stipulation, and the distinction is load-bearing. The roundtrip tells you when you have succeeded. The five conditions tell you what to check, and they decompose the requirement into parts that turn out to have very different relationships to the machinery we have.

*Remark 4.6* (What the framework does and does not do). Goodman’s conditions are conditions on a symbol *system*: a scheme together with a compliance relation. They presuppose the compliance relation as given and adjudicate whether it is notational. They therefore cannot *select* a compliance relation, and they are invariant under relabelling: if  $\Sigma$  is a notational system and  $\phi$  is a bijection on characters,  $\phi(\Sigma)$  is a notational system with the same compliance classes differently named. Consequently Goodman’s framework says what a solution to the bootstrap problem would look like, and cannot by itself produce one.

Remark 4.6 is the reason this note describes the adoption of Goodman as “half the battle” rather than as progress on the bootstrap problem proper. It returns with force in §7.

#### 4.4 Failure modes and provisional repairs

**The wrong-note paradox.** Goodman’s compliance relation is exact: a performance containing a single wrong note is not a performance of the work. Since performances without wrong notes are essentially never found, the theory as stated has no instances in musical practice. This is the standard objection and it is a real one.

We record the shape of the repair now and return to it in §5.6: exactness should be relativised to a resource bound. Two performances are the same performance if no affordable observation separates them. This is not an evasion, because in a cost-accounted setting “affordable” is a defined quantity rather than a hand-wave, and it converts an absolute condition into a family of conditions indexed by budget.

**Density is a property of the world, not only of the notation.** Finite differentiation can fail because the notation is badly designed, or because the phenomenon being notated is dense and no notation of it can succeed. Goodman does not clearly separate these, and for the bootstrap problem the separation is essential: we need to know whether a failure to notate is a failure of the agent or a fact about the environment. See Question 2.

**The ontological commitments are separable.** Goodman’s theory of notation comes attached to strong claims about the identity of works of art. Recent work in aesthetics argues that the notational apparatus survives being detached from those commitments and reinterpreted epistemically [9]. We take that detachment for granted here.

### 5 Fit with a bisimulation-based approach

#### 5.1 The two bisimulations

The natural translation of the roundtrip requirement into process-calculus vocabulary is a pair of bisimulations. Write  $\mathcal{S}$  for the space of scores,  $\mathcal{P}$  for the space of performances,  $\llbracket - \rrbracket : \mathcal{S} \rightarrow \mathcal{P}(\mathcal{P})$  for the compliance map taking a score to its compliance class, and  $\tau : \mathcal{P} \rightarrow \mathcal{S}$  for transcription. The requirement is that

$$p \approx_{\mathcal{P}} p' \iff \tau(p) \approx_{\mathcal{S}} \tau(p') \tag{1}$$

$$s \approx_{\mathcal{S}} s' \iff \llbracket s \rrbracket = \llbracket s' \rrbracket \tag{2}$$

with  $\approx_{\mathcal{P}}$  and  $\approx_{\mathcal{S}}$  the respective behavioural equivalences. Read one way this says the two kernels agree; read another, that  $\tau$  and  $\llbracket - \rrbracket$  are mutually inverse on the quotients.

## 5.2 What bisimulation supplies

Conditions 4.1, 4.3 and 4.4 are quotient conditions and fall out of (1)–(2) directly. Syntactic disjointness says the character map is a function on marks, which is what it means for  $\approx_{\mathcal{S}}$  to be an equivalence. Unambiguity says the compliance class of a character does not vary, which is a statement that  $\llbracket - \rrbracket$  is well defined on  $\approx_{\mathcal{S}}$ -classes. Semantic disjointness says compliance classes partition, which is (2) read as injectivity on the quotient.

This is a good fit, and it is why the translation is worth making at all. Bisimulation is the mathematically mature version of “same behaviour”, it comes with coinductive proof methods, and in the reflective setting it is the equivalence the logic already denotes.

## 5.3 Where Goodman adds

Conditions 4.2 and 4.5 are *not* quotient conditions and bisimulation does not supply them.

**Observation 5.1** (Separation is not quotient). *Finite differentiation is a separation condition: it requires not merely that distinct characters be distinct, but that their distinctness be effectively determinable from a mark. A bisimulation quotient can be perfectly well defined over a dense space, in which case every pair of distinct classes is distinguishable in the limit and no pair is distinguishable by any finite observation. Such a system satisfies (1)–(2) and fails Condition 4.2.*

This is the precise content of “Goodman is compatible with the bisimulation approach but adds more conditions”. What he adds is an effectivity requirement that process equivalence, being defined by a greatest fixed point, is systematically silent about.

The point is not academic. It is exactly where the neural implementations fail.

## 5.4 Two empirical instances of the gap

**LatPlan and the symbol stability problem.** LatPlan learns propositional and action symbols from unlabelled image pairs and plans in the resulting latent space [2]. Asai and Kajino subsequently identified what they named the *symbol stability problem*: the propositions produced by the state autoencoder behave stochastically, so the same input does not reliably produce the same proposition [3]. In Goodman’s vocabulary this is a failure of Condition 4.2—one mark, no effective determination of which character it belongs to—and it was discovered empirically, without the vocabulary, as an obstacle to making a learned symbol system behave like a symbol system.

**Vector quantisation as a fix for one condition only.** The VQ-VAE bottleneck assigns continuous latents to a codebook by nearest neighbour [18]. This imposes syntactic disjointness and a separation margin by construction: it is, read through Goodman, an engineering solution to Conditions 4.1 and 4.2 and to nothing else. It says nothing about the compliance side, which is why quantisation alone does not produce symbols that mean anything. The subsequent literature’s efforts to make the codes “semantic” by additional training objectives are attempts to buy Conditions 4.3–4.5 that do not know they are doing so.

## 5.5 Why bisimulation metrics are the wrong repair

The reinforcement-learning literature has independently encountered the rigidity of exact bisimulation and responded with quantitative relaxations: bisimulation metrics [10] and their

scalable descendants [11, 26, 8]. These are the correct response to the wrong-note paradox in that setting, and they are the wrong response here.

**Observation 5.2** (The metric smooths what notation needs separated). *A bisimulation metric replaces the equivalence with a distance and the quotient with a geometry. It thereby dissolves precisely the discreteness that Conditions 4.2 and 4.5 require. A representation shaped by a bisimulation metric is, in Goodman’s classification, closer to a sketch than to a score.*

The tension is genuine and not merely terminological. One cannot have a symbol system and a smooth behavioural geometry at the same level of description. What one can have is a discrete system whose *admissible discriminations* are set by a resource bound, which is the next subsection.

## 5.6 Budget-relative separation

The mortal-scientist framework already prices observation: assays cost, refutation is budget-relative, and a verdict may be  $\perp$  because the budget ran out rather than because the world was silent. This supplies exactly the ingredient Observation 5.1 says is missing.

**Conjecture 5.3** (Finite differentiation as a theorem). *In a cost-accounted setting with budget  $\sigma$ , define  $x \approx^\sigma y$  to hold when no assay affordable at  $\sigma$  separates  $x$  from  $y$ . Then the quotient by  $\approx^\sigma$  satisfies Conditions 4.2 and 4.5 with the budget as the separation constant, and the resulting family  $\{Q_\sigma\}_\sigma$  is a filtered diagram of notational schemes refining as  $\sigma$  grows.*

We state this as a conjecture rather than a proposition because two things are unverified. First, that  $\approx^\sigma$  is transitive: affordability is not obviously composable, and a chain of pairwise-indistinguishable states may have distinguishable endpoints, which is the standard failure of “indistinguishable by a bounded observer” relations. Second, that the cost-accounting monad interacts correctly with the extension of §6; see Question 6. If the first fails, the repair is to take the transitive closure and accept a coarser scheme, at the price of the separation constant no longer being  $\sigma$ .

Should Conjecture 5.3 hold, it has an appealing reading: a poorer agent has a coarser notation, notational adequacy is a purchasable property, and the wrong-note paradox is resolved by making exactness relative to what can be afforded rather than by abandoning it.

## 6 rho and builtin processes

### 6.1 Reflection is score construction, not performance

The reflective higher-order calculus takes names to be quoted processes [17]. For a process  $P$ ,  $@P$  is a name; for a name  $x$ ,  $*x$  is a process; and  $*@P = P$ ,  $@*x = x$ . Quotation and dereference are mutually inverse.

It is tempting—and we made this mistake before correcting it—to read this immediately as the score / performance relation with Goodman’s roundtrip holding by construction. It is not, and seeing why is what makes the rest of this section follow rather than intrude.

In pure rho, both sides of  $@$ — and  $*$  are syntax.  $P$  is a term;  $@P$  is a term-code;  $*@P$  is the same term again. Nothing here is a performance. What the operators do is build scores out of scores: quotation lets a score be embedded in another score as an atom, and dereference recovers it. The construction is *recursive score construction*, and its losslessness is the statement that the recursion is faithful—that no information is lost by descending a level and climbing back. That is a genuine and useful property, but it is a property internal to

the notation. A notational system all of whose compliants are further inscriptions is exactly the case Goodman analyses under the heading of scripts that denote other scripts, and it is not yet a system that says anything about a world.

**Observation 6.1** (Two readings of  $@-$  and  $*$ ). *In  $\rho = \text{RHO}[\emptyset]$ ,  $@-$  and  $*$  are score-to-score: they witness the faithfulness of a recursion inside the notation, and there is no performance in the picture. The compliance relation is trivial in the sense that every character complies with syntax.*

## 6.2 Builtins turn the recursion into a correspondence

Now adjoin a builtin process  $b$  drawn from the host world. The syntax of quotation does not change:  $@b$  is formed exactly as  $@P$  is. But its *semantics* changes completely, because  $b$  is not a term of the calculus. It is a behaviour in the world the computation is embedded in.

At that point, and only at that point,  $@b$  is a score and  $b$  is a performance in Goodman's sense.  $@b$  is a static, inspectable, transmissible mark that can be handled by  $\rho$ 's own machinery—sent on channels, pattern-matched, quantified over in the namespace logic—while the thing it codes is an event elsewhere, running under laws the calculus does not supply. Dereference  $*@b$  is then not a syntactic identity but a performance instruction: it says, produce in the host world something indistinguishable from  $b$ .

**Observation 6.2** (Where the correspondence enters). *Adjoining builtins does not add a new operator. It re-types the existing one. The same  $@-$  that was recursive score construction over  $\rho$ -terms becomes, on the builtin sort, a score / performance correspondence; and the same  $*$  that was syntactic recovery becomes performance. Goodman's roundtrip requirement therefore attaches to  $\rho$  at exactly the point where the calculus is extended by the world, and nowhere else.*

This is why the extension is the interesting object and the pure calculus is not. In  $\text{RHO}[\emptyset]$  there is nothing to ground and the roundtrip is a tautology. In  $\text{RHO}[\mathcal{B}]$  the roundtrip is a claim about the relation between a term language and a world, which is the kind of claim the bootstrap problem is about. Everything the rest of this section does is take that observation seriously and see how far it can be pushed—and §6.6 argues that the answer is: not far enough.

## 6.3 Two ways to adjoin the world

There are two ways to perform the extension, and Observation 6.2 explains why only one of them does the job.

**Builtin names.** Adjoin name constants beyond  $@0$ , understood as channels to the host world. This gives an interface but no theory of what is on the other end: the calculus can send and receive, and has nothing to say about the behaviour it is interacting with. It is the applied- $\pi$  move [1], where a term algebra of data is adjoined and the added material is inert with respect to the operational semantics. Crucially, it does *not* re-type  $@-$ : an adjoined name is a name, not the quotation of anything, so there is no performance for it to be the score of, and Observation 6.1 still applies. The roundtrip stays inside the notation.

**Builtin processes.** Adjoin processes  $b$  drawn from the host world. Then  $@b$  is a *symbol* for  $b$ , and  $*@b$  is a process in the host world indistinguishable from  $b$ . This is the version that does the work, for the reason just given: it is the only one of the two under which quotation acquires a compliance class outside the syntax.

*Remark 6.3* (This is not psi-calculi). The psi-calculi framework parameterises  $\pi$ -like calculi by nominal datatypes for terms, conditions and assertions, together with a channel-equivalence relation, a composition, a unit and an entailment relation, and subsumes applied  $\pi$ , spi, fusion and concurrent constraint  $\pi$  as instances [6]. The adjoined material there is *data*: inert under the operational semantics, transmitted but not run. Adjoining builtin *processes* is a different move, because the adjoined objects have their own dynamics and the calculus ceases to be closed—part of the reduction relation is delegated to the host. Readers who know psi-calculi will otherwise assume this is an instance of it.

## 6.4 The calculus-making functor

The natural way to organise this is to stop thinking of rho as a calculus and think of it as a functor from behaviours to calculi:

$$\text{RHO}[\mathcal{B}] ::= 0 \mid \mathcal{B} \mid \text{for}(y \leftarrow x)P \mid x!(Q) \mid P \mid Q \mid *x, \quad x, y ::= @P$$

with the usual equations and rewrites, and with  $\mathcal{B}$ 's behaviour specified in a source category  $\mathbf{C}$  and imported into a version of the parallel-composition rule. The pure calculus is then the fibre over the empty behaviour,  $\text{rho} = \text{RHO}[\emptyset]$ , and everything previously done with rho has been done in that fibre.

## 6.5 What the functorial reading suggests

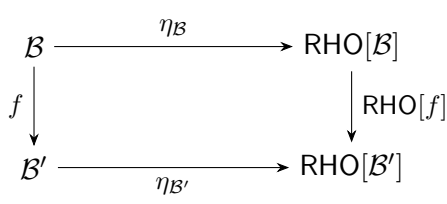
Four things, each of which is a suggestion rather than a theorem.

(i) **The import into  $\mid$  has a known shape.** “Behaviour specified in the source category, imported into the par rule” is the shape of a distributive law of syntax over behaviour. Turi and Plotkin’s bialgebraic semantics takes operational rules given as a natural transformation  $\lambda : SF \Rightarrow FS^\dagger$ , with  $S$  a syntax functor,  $F$  a behaviour functor and  $S^\dagger$  the free monad on  $S$ , and yields an operational model together with a canonical, internally fully abstract denotational model [25]. The relevant corollary is that for any such law, behavioural equivalence in the final coalgebra is a congruence on the operational model. If the import can be put in that form, bisimulation-as-congruence over  $\text{RHO}[\mathcal{B}]$  comes free rather than being re-proved per  $\mathcal{B}$ . If it cannot, congruence fails and every compositional result of the existing notes stops transferring across the cut. This is checkable and unchecked.

(ii) **Functoriality is the roundtrip.** Take the morphisms of  $\mathbf{C}$  to be behaviour-preserving maps. There is then a unit  $\eta_{\mathcal{B}} : \mathcal{B} \rightarrow \text{RHO}[\mathcal{B}]$  including builtins as processes, and:

- $\eta$  faithful says distinct builtin behaviours receive distinct symbols, which is performance  $\rightarrow$  score  $\rightarrow$  performance losslessness;
- $\eta$  full says every rho-level distinction over builtins is realised by a builtin distinction, which is score  $\rightarrow$  performance  $\rightarrow$  score losslessness, and which forbids the calculus from minting symbols the world cannot cash.

So Goodman’s criterion of adequacy becomes the statement that  $\eta_{\mathcal{B}}$  is full and faithful with respect to behavioural equivalence. That is a property one could try to prove, which is more than the criterion was before.



$\eta$  faithful: no two behaviours share a symbol.

$\eta$  full: no symbol without a behaviour.

Naturality: refinement of the world lifts to refinement of the calculus.

Figure 2: The functorial reading. Goodman’s two-way losslessness becomes fullness and faithfulness of the unit; extensibility of the builtin set becomes naturality in  $\mathcal{B}$ .

(iii) **Unambiguity forces behaviours, not states.** If the objects of  $\mathcal{B}$  are states with dynamics, then  $@b$  codes a moving target: the same mark has different compliants at different times, which is exactly the failure of Condition 4.3. The repair is forced: the objects of  $\mathcal{B}$  must be *behaviours*—a subobject of the final  $F$ -coalgebra—so that quoting a builtin quotes its entire future rather than a snapshot. Then  $*@b \approx b$  holds definitionally, and unambiguity holds because behaviours do not change over time even though states do. A consequence worth stating: one cannot take  $\mathcal{B}$  to be the physical furniture of the world.  $\mathcal{B}$  must be the world already quotiented by an observational equivalence, which is where Conjecture 5.3 would enter, with  $\mathcal{B} = Q_{\sigma}(W)$ .

(iv) **The source category is constrained.** Name matching in  $\text{for}(y \Leftarrow x)P \mid x!(Q)$  requires deciding name equality, and names now include  $@b$ . So structural congruence on  $\text{RHO}[\mathcal{B}]$  is decidable only if equality on  $\mathcal{B}$ ’s objects is at least budget-decidable. This is a restriction on which worlds can serve as source, and it is the formal descendant of the observation that builtin names are the base case of the well-founded coding by which quotation manufactures fresh names.

## 6.6 Why this is an analogy and not a solution

We now argue against the construction just presented, because the argument is short and decisive and it is better made here than by a reader.

- (a) **It consumes what it was meant to produce.**  $\text{RHO}[-]$  takes  $\mathcal{B}$  as an argument. The bootstrap problem is the problem of *finding*  $\mathcal{B}$ : of individuating the world into behaviours worth having symbols for. The functor presupposes a solved instance of the problem it was introduced to address, and everything it then does is bookkeeping over that solution. This is the decisive objection and the rest are supporting.
- (b) **Functoriality entails relabelling invariance.**  $\text{RHO}[\mathcal{B}] \cong \text{RHO}[\mathcal{B}']$  whenever  $\mathcal{B} \cong \mathcal{B}'$ . So the construction cannot fix reference; it transports whatever reference  $\mathcal{B}$  came with. Note that this is not an imported objection but a theorem of the construction, and it is the formal version of Remark 4.6.
- (c) **Losslessness is stipulated where it is interesting.**  $*@b \approx b$  holds by construction because we defined  $@-$  on  $\mathcal{B}$  to be exact. In the world, sensor-to-symbol transduction is lossy, noisy and partial, and the interesting question is what to do when the roundtrip *fails*. The construction has nothing to say about approximate compliance, which is where all the empirical difficulty lives.

- (d) **The par rule import is unverified.** §6.5(i) is conditional. Until the import is exhibited as a distributive law of the required shape, congruence of bisimulation over  $\text{RHO}[\mathcal{B}]$  is an assumption, and with it every compositional result that would be transported.
- (e) **Two interaction disciplines complicate the logic.** rho communication is name-mediated. If builtins react in the source category by something that is not a name rendezvous,  $\text{RHO}[\mathcal{B}]$ 's parallel composition hosts two kinds of redex, and the namespace logic's modalities—which quantify over redex positions—require a sort distinction. The ladders computed in the existing notes would need recomputation, and it is not known whether their qualitative shape survives.
- (f) **Nothing here proposes candidates.** Even granting all of the above, the construction is a specification language for grounded symbol systems, not a procedure for acquiring one. It says what the answer must look like. It does not search.

The honest summary is that the functorial reading is a good way to *state* several of Goodman's conditions in a form where they might be proved, and no way at all to satisfy them.

## 7 Reference cannot be fixed from inside

One consequence of §6.6(b) deserves separate statement, because it changes the shape of the project rather than merely qualifying it.

Suppose an agent solves the bootstrap problem in the sense stated: it constructs a symbol system, its own, notationally adequate, with correspondence to its environment instantiated in itself. Nothing in that achievement makes the system correspond to *our* symbols. By Remark 4.6 and §6.6(b), the system is determined up to isomorphism and no further. A grounded but idiosyncratic symbol system is Rongorongo viewed from the other side: perfectly meaningful to its users, opaque to everyone else, and opaque for exactly the reason given in Observation 2.1.

**Observation 7.1** (The social term is forced). *For the target “a mind that acts in the world humans act in”, grounding is a property of a composite of at least two agents with shared action, not of any single agent. Any framework that is invariant under relabelling of a single agent's symbols cannot deliver the target, and  $\text{RHO}[-]$  is such a framework by construction.*

This is the point at which the emergent-communication literature stops being adjacent and starts being load-bearing. It is also where the argument of this section meets the existence proofs of §2.6 coming the other way: relabelling-invariance says a single agent *cannot* fix reference, and the only two known solutions to the bootstrap problem were both achieved by populations and by neither lineage's individuals. A structural argument and an empirical one, with no premises in common, reaching the same requirement. Steels's language-game experiments demonstrate shared vocabulary arising between embodied agents through repeated situated interaction [23], and Taniguchi's collective predictive coding hypothesis reconstructs the phenomenon as decentralised Bayesian inference over latent variables that are common nodes connecting many agents, achievable without connecting brains [24]. Neither solves the bootstrap problem—both assume pre-segmented perceptual channels, which is objection §6.6(a) again—but both supply the ingredient a single-agent account structurally cannot.

For the present programme the practical consequence is that *Composing Learners* is not an extension of the mortal-scientist framework but a prerequisite for the grounding question. A learner as a distribution over a population, composed by parallel composition with typed namespace overlap, is the right shape of object to carry a correspondence that no member of the population carries alone.

## 8 Where a transformer belongs

The framing above assigns transformers a job, and it is not the job they are usually given in discussions of grounding.

In the construction of §6 the transducer is @– and \* extended to builtins. That map is exact and lossless by construction; a network cannot improve on it and is not needed for it. What is needed, and what nothing in the calculus supplies, is a *proposal mechanism*: given a stream of host-world observation, propose candidate builtin behaviours. This is objection §6.6(a) turned into a task.

Two features make it a well-posed learning problem, which is more than can be said for “learn symbols from sensors”.

**The success criterion is coverage, not reward.**  $\mathcal{B}$  is a *presentation*: a generating set such that the observable world lies in the image of  $\text{RHO}[\mathcal{B}]$ . This is a generators-and-relations problem with a definite answer, and minimality of  $\mathcal{B}$  is a statistical-complexity analogue rather than a hyperparameter.

**There is a principled candidate.** Causal states are already the coarsest coarse-graining of pasts retaining full predictive power [21], and by §2.4 a next-token-trained transformer represents belief states over them [20]. That makes causal states the natural first candidate for which distinctions deserve builtin names. The correspondence extends further than convenience: Crutchfield’s taxonomy of  $\epsilon$ -machines runs from finite-state through countable to uncountable and fractal, and that taxonomy *is* a density hierarchy—so Condition 4.2 is the condition separating its bottom rung from the rest, and “is this world notationally accessible to this agent at this budget” becomes “is the budget-quotiented  $\epsilon$ -machine of the host process finite”.

The second job for a network is amortised bisimulation checking. Assays are priced in the existing framework, checking behavioural equivalence is expensive, and an approximate checker with calibrated error is a legitimate purchase if its errors can be priced. This is more speculative and we do not pursue it here.

## 9 Open questions

**On the notational apparatus.**

*Question 1* (Transitivity of budget-relative indistinguishability). Is  $\approx^\sigma$  of Conjecture 5.3 transitive? If not, what is the cost of taking transitive closure, and does the resulting scheme still separate at a rate related to  $\sigma$ ?

*Question 2* (Density: agent or world?). Given a failure of Condition 4.2, is there a criterion separating “this agent cannot notate this world” from “this world admits no notation”? A candidate: the former is a statement about  $Q_\sigma$  for the agent’s  $\sigma$ , the latter about  $\lim_{\sigma \rightarrow \infty} Q_\sigma$ .

*Question 3* (Notationality of learned codebooks). Goodman’s five conditions are checkable properties of an encoder / decoder pair. What do VQ-VAE codebooks, BPE tokenisers and LatPlan’s state autoencoder score on them? “How notational is this tokeniser” appears to be a measurable quantity and, so far as we can determine, has never been measured.

**On the functor.**

*Question 4* (Is the import a distributive law?). Can the import of  $\mathcal{B}$ ’s behaviour into parallel composition be exhibited as  $\lambda : \text{RHO} \circ F \Rightarrow F \circ \text{RHO}^\dagger$ ? If so, congruence is free by [25]. If not, what weaker condition holds, and does the lax version suffice?

*Question 5* (Is  $\mathbf{RHO}$  a monad?). Does  $\mathbf{RHO}[\mathbf{RHO}[\mathcal{B}]] \rightarrow \mathbf{RHO}[\mathcal{B}]$  flatten? If so its algebras are the worlds closed under rho-computation, which would be a sharp notion of “a world that can host”. Note that abstract GSOS wants the syntax as a monad or its free monad, so this question and Question 4 are coupled.

*Question 6* (Does  $\mathbf{RHO}$  preserve the cost-accounting monad?). The existing notes work inside plain rho and appeal to a WLOG on syntax that is valid there. If cost accounting is a monad  $T$ , do  $\mathbf{RHO}[-]$  and  $T$  commute? If they do not, the foraging inequality, the pricing schedule and the monotonicity of cost in distance do not transport to  $\mathbf{RHO}[\mathcal{B}]$ , and with them Conjecture 5.3 loses its setting.

*Question 7* (Full and faithful  $\eta$ ). For which source categories  $\mathbf{C}$  and which subobjects  $\mathcal{B}$  of the final coalgebra is  $\eta_{\mathcal{B}}$  full and faithful with respect to weak bisimulation? Failure of fullness would be the calculus minting uncashable symbols, which is a diagnosable pathology and possibly a measurable one.

*Question 8* (Sorting the logic). If  $\mathbf{RHO}[\mathcal{B}]$  hosts two kinds of redex, what is the corresponding refinement of the namespace logic’s modalities, and does the qualitative shape of the ladders computed in *Learning to Play* and *Composing Learners* survive it?

### On the bootstrap proper.

*Question 9* (Proposing  $\mathcal{B}$ ). Given host-world observation, what proposes candidate builtin behaviours, and against what loss? §8 argues the criterion is coverage of the observable world by  $\mathbf{RHO}[\mathcal{B}]$ . Making that a differentiable or otherwise optimisable objective is open.

*Question 10* (Extensibility). Acquiring a new distinction should be a morphism  $\mathcal{B} \rightarrow \mathcal{B}'$  in  $\mathbf{C}$  that  $\mathbf{RHO}$  transports. What is the well-founded discipline under which an agent may extend its own builtin set—presumably by quotation over existing builtins, which is the algebraic form of generate-and-test feature construction in the sense of [19]?

*Question 11* (The minimal social configuration). Observation 7.1 says grounding is a property of a composite. What is the smallest composite that suffices? Two agents with shared action, or does the argument require a population? Does the typed-overlap apparatus of *Composing Learners* already contain the answer?

*Question 12* (Approximate compliance). §6.6(c) observes that the interesting case is roundtrip failure. Is there a notion of *graded* compliance that degrades Goodman’s conditions gracefully without collapsing into the metric of Observation 5.2?

## 10 What is claimed and what is not

**Claimed.** That the correspondence between a language and a world is a fact about the language’s users and is not contained in the corpus (Observation 2.1); that the record of decipherment and of cetacean bioacoustics is evidence for this rather than merely consistent with it, and that the two successful routes both acquire the correspondence from agents who already have it (Observation 2.2); that next-token prediction recovers the structure of the token-generating process, which for a corpus is the linguistic community and not the environment (Observation 2.3); that the problem has at least two independent solutions in nature, so that solubility is not in question and impossibility arguments are unsound (Observation 2.4); that the mortal-scientist framework’s grounding is analytic and therefore does not address this (Observation 3.1); that Goodman’s conditions decompose into quotient conditions that bisimulation supplies and separation conditions that it does not (Observation 5.1); that bisimulation metrics are the wrong repair for the latter (Observation 5.2); that quotation and dereference in the pure calculus are score-internal recursion and acquire a compliance

class outside the syntax only when builtin processes are adjoined (Observations 6.1 and 6.2); and that no relabelling-invariant single-agent framework can deliver correspondence to *human* symbols (Observation 7.1).

**Conjectured.** That cost accounting supplies finite differentiation with the budget as separation constant (Conjecture 5.3).

**Offered as analogy only.** The entire content of §6. The functorial reading organises the conditions attractively and suggests four checkable statements. It does not bootstrap anything, for the reasons given in §6.6, and the first of those reasons is not a technicality.

**Not claimed.** That Goodman’s conditions are the right ones—they are adopted provisionally, and the density and wrong-note objections are live. That transformers cannot in principle transduce—only that next-token prediction over an already-grounded corpus is not transduction, and that the two are routinely conflated. That the bootstrap problem is soluble *by construction*: nature’s two solutions were achieved by populations, under selection, over long timescales, with interfaces that co-evolved with the symbol systems they support, and this note has no argument that any of those conditions can be dispensed with.

## References

- [1] M. Abadi and C. Fournet. Mobile values, new names, and secure communication. *POPL*, 2001.
- [2] M. Asai and A. Fukunaga. Classical planning in deep latent space: bridging the subsymbolic-symbolic boundary. *AAAI*, 2018.
- [3] M. Asai and H. Kajino. Towards stable symbol grounding with zero-suppressed state autoencoder. *ICAPS*, 2019.
- [4] E. M. Bender and A. Koller. Climbing towards NLU: on meaning, form, and understanding in the age of data. *ACL*, 2020.
- [5] G. Beguš, M. Dąbkowski, R. L. Sprouse, D. F. Gruber and S. Gero. The phonology of sperm whale coda vowels. *Proceedings of the Royal Society B* 293(2069):20252994, 2026.
- [6] J. Bengtson, M. Johansson, J. Parrow and B. Victor. Psi-calculi: a framework for mobile processes with nominal data and logic. *Logical Methods in Computer Science* 7(1), 2011.
- [7] Y. Bisk, A. Holtzman, J. Thomason, J. Andreas, Y. Bengio, J. Chai, M. Lapata, A. Lazaridou, J. May, A. Nisnevich, N. Pinto and J. Turian. Experience grounds language. *EMNLP*, 2020.
- [8] P. S. Castro, T. Kastner, P. Panangaden and M. Rowland. MICO: improved representations via sampling-based state similarity for Markov decision processes. *NeurIPS*, 2021.
- [9] On the theory of musical scores and its relation to work-performance. *Estetika: The European Journal of Aesthetics*, 2026.
- [10] N. Ferns, P. Panangaden and D. Precup. Metrics for finite Markov decision processes. *UAI*, 2004.
- [11] C. Gelada, S. Kumar, J. Buckman, O. Nachum and M. G. Bellemare. DeepMDP: learning continuous latent space models for representation learning. *ICML*, 2019.

- [12] V. Goel. Notationality and the information processing mind. *Minds and Machines* 1(2):129–166, 1991.
- [13] N. Goodman. *Languages of Art: An Approach to a Theory of Symbols*. Bobbs-Merrill, 1968; second edition Hackett, 1976.
- [14] S. Harnad. The symbol grounding problem. *Physica D* 42:335–346, 1990.
- [15] J. Luo, Y. Cao and R. Barzilay. Neural decipherment via minimum-cost flow: from Ugaritic to Linear B. *ACL*, 2019.
- [16] J. Luo, F. Hartmann, E. Santus, Y. Cao and R. Barzilay. Deciphering undersegmented ancient scripts using phonetic prior. *TACL*, 2021.
- [17] L. G. Meredith and M. Radestock. A reflective higher-order calculus. *Electronic Notes in Theoretical Computer Science* 141(5):49–67, 2005.
- [18] A. van den Oord, O. Vinyals and K. Kavukcuoglu. Neural discrete representation learning. *NeurIPS*, 2017.
- [19] D. Pierce and B. J. Kuipers. Map learning with uninterpreted sensors and effectors. *Artificial Intelligence* 92:169–229, 1997.
- [20] A. S. Shai, S. E. Marzen, L. Teixeira, A. Gietelink Oldenziel and P. M. Riechers. Transformers represent belief state geometry in their residual stream. *NeurIPS*, 2024.
- [21] C. R. Shalizi and J. P. Crutchfield. Computational mechanics: pattern and prediction, structure and simplicity. *Journal of Statistical Physics* 104:817–879, 2001.
- [22] P. Sharma, S. Gero, R. Payne, D. F. Gruber, D. Rus, A. Torralba and J. Andreas. Contextual and combinatorial structure in sperm whale vocalisations. *Nature Communications* 15:3617, 2024.
- [23] L. Steels. *The Talking Heads Experiment: Origins of Words and Meanings*. Language Science Press, 2015.
- [24] T. Taniguchi. Collective predictive coding hypothesis: symbol emergence as decentralized Bayesian inference. *Frontiers in Robotics and AI* 11:1353870, 2024.
- [25] D. Turi and G. D. Plotkin. Towards a mathematical operational semantics. *LICS*, 1997.
- [26] A. Zhang, R. McAllister, R. Calandra, Y. Gal and S. Levine. Learning invariant representations for reinforcement learning without reconstruction. *ICLR*, 2021.